Converting PFAS Dissociation Constants to Protein–Water Partition Coefficients

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Summary

A PFAS dissociation constant converts to a protein-water partition coefficient exactly, in one line of arithmetic, and the conversion is still not worth much. Four findings, in the order they matter.

1. **The conversion is exact but the constant is protein-specific.** `log K = log10(1e9/M) + log n - log Kd`. The leading term is 4.1774 for albumin and 4.8475 for liver FABP. Using albumin's on an L-FABP constant is 4.7x low, silently, with no diagnostic in the output.
2. **Converted values sit 0.80 log below measured ones**, consistently, over the 22 pairs from our best-powered source. That is a missing site count: closing it needs about six sites where a single-site fit allowed one.
3. **Non-specific binding is not the explanation.** We tested it. At the trace limit the specific term carries 86-98% of the total, so dropping the non-specific term costs 0.07 log at most - twelve times too small.
4. **The input is noisier than any of this.** Two laboratories measuring the same albumin Kd by the same method differ by 1.45 log, 45x on PFOA. A converted log K inherits that one for one, so it cannot be better than 28x reproducible whatever stoichiometry anyone assumes.
5. **Species barely matters here and sex cannot.** Albumin constants move 0.09 log between human and bovine and 0.36 log across four mammals, both below the 0.59 log between-laboratory floor. No binding row resolves sex, because purified albumin has none. The large species and sex effects in PFAS toxicokinetics are in transport, not binding.

The practical consequence: use a measured partition coefficient where one exists, and treat a converted one as an order-of-magnitude estimate of the specific term alone.

Scope: the ten sources that supply the most measured constants against a named single protein. Four findings rest on papers outside that set; those are named at the end.

The two quantities, and three traps

A dissociation constant and a partition coefficient answer different questions about the same experiment, which is why one converts into the other at all.

Kd is the free concentration at which half the sites are occupied. Units of concentration. It describes **one class of site** and says where that site saturates.

K (or D) is the ratio of PFAS in the protein phase to PFAS in water, at equilibrium. Units of volume per mass (or per volume) of protein. It describes **everything the protein holds** and, being a ratio, is a constant only while nothing is saturating.

The same number can be written either way. Three things stop them being interchangeable in practice, and all three have bitten this dataset.

Trap	What goes wrong	Size
Basis	per kg of protein vs per litre of protein	0.135 log (1.364x, from albumin's 0.733 mL/*g* partial specific volume)
Sorbent definition	protein alone vs protein + lipid vs whole serum	unbounded; a whole-matrix coefficient is what a mass balance must *predict*, not an input to one
Speciation	K is one species, D is all species at a stated pH	negligible for the acids; material only for PFOSA (pKa 8.77, 4.1% anionic at pH 7.4)

Fischer 2024 and Allendorf 2019 report per kg; Qin 2024 and Bischel 2011 report per litre. Every number in this document is on the **per-kg** basis, with the density conversion applied where the source used the other one.

The derivation, in full

Five steps. The important one is step 3, where the protein concentration cancels.

Step 1 - site occupancy. For one class of site with total protein P and free ligand Cw, the bound complex PL obeys the mass-action definition of Kd:

```
K_d = \frac{[P][C_w]}{[PL]}
```

With the site balance `P_total = [P] + [PL]`, the free protein `[P]` cancels and the fraction of sites occupied is

```
\theta = \frac{C_w}{K_d + C_w}
```

Step 2 - bound amount. If each protein molecule carries n such sites, the ligand bound per litre of the protein phase is

```
C_{bound} = P \cdot n \cdot \theta = \frac{P \, n \, C_w}{K_d + C_w}
```

Step 3 - divide by the protein mass, and watch P vanish. The partition coefficient is bound ligand per kilogram of protein, divided by the ligand left in water. One litre holding P micromoles of protein of molar mass M weighs `P x M / 1000` grams, so

```
K = \frac{C_{bound}}{C_w} \cdot \frac{1000}{P \, M} = \frac{1000 \, n}{M \, (K_d + C_w)}
```

P has cancelled. This is the whole reason the conversion works: a partition coefficient does not depend on how much protein was in the cell, which is exactly what makes it transferable between an assay and a tissue. It does depend on Cw, which is what people forget.

Step 4 - the trace limit. As `Cw -> 0`, which is where a partition coefficient is a constant at all:

```
K^0 = \frac{1000 \, n}{M \, K_d}
```

Step 5 - put Kd in micromolar and take logs. With Kd in uM rather than molar, the 1000 becomes 1e9:

```
\log K = \log_{10}\!\left(\frac{10^9}{M}\right) + \log n - \log K_d(\mu M)
```

The `1e9` and not `1e6` is what makes the two sides commensurate. Getting it wrong is a silent factor of 1000, and it put every implied site count in an early draft of our code between 156 and 117,000.

The leading term is not a fudge factor. `1e9/M` is the number of micromoles of protein present in one kilogram of that protein - 15,046 for albumin. It is what puts bound amount (mol per mol of protein) and partition coefficient (litres of water per kg of protein) in the same units.

The constant is protein-specific

This is the one that will bite people, because 4.1774 looks like a universal constant and is not. A lighter protein packs more binding sites into a kilogram, so the **same Kd gives a larger partition coefficient**.

Masses are the mature chain computed from the UniProt sequence (average residue masses, water added, two hydrogens removed per disulfide), which is what a purified preparation actually weighs.

Protein	M (g/mol)	umol per kg	log10(1e9/M), per kg	per litre
Serum albumin (HSA/BSA)	66,463	15,046	**4.1774**	4.3123
Human L-FABP (P07148)	14,208	70,383	**4.8475**	4.9824
Human I-FABP (P12104)	15,106	66,199	4.8209	4.9558
Rat L-FABP (P02692)	14,272	70,067	4.8455	4.9804

Albumin to human L-FABP is **+0.670 log, a factor of 4.68**.

Worked, on the lightest protein we hold. PFOA on human L-FABP by equilibrium dialysis, single-site fit, Kd = 0.099 uM:

```
\log K = 4.8475 + \log_{10}(1) - \log_{10}(0.099) = 5.85
```

Using albumin's constant by mistake gives **5.18** instead. Both look like reasonable log K values and nothing in the number says which is right.

Species differences inside albumin do not matter. HSA computes to 66,437 and BSA to 66,398 - 0.00025 log apart, three orders below every other uncertainty here. One mass is used for both.

What we cannot convert, and why it is instructive. Seven PPAR constants have an affinity *and* a stoichiometry and still cannot be converted: the source names only the supplier, never whether the material is the full-length receptor (52.2, 57.6 and 49.9 kDa for alpha, gamma, delta) or the ligand-binding domain alone, about a third of that. A factor of three in M is 0.48 log. The missing input is the **mass of the thing that was weighed**, not the stoichiometry.

Two corrections

Several classes of site

Step 2 of the derivation summed over one class. Several classes simply add:

```
K^0 = \frac{10^9}{M} \sum_i \frac{n_i}{K_{d,i}}
```

Worked on the one source in our set that fits PFOA on rat L-FABP twice, by two methods:

Fit	Kd values (uM)	log K0
Displacement, 2 site classes	13.1, 23.8	3.92
ITC, 3 site classes	3.1, 26.2, 52.6	4.43

The two differ by 0.51 log (3.2x). Read that carefully, because it is the whole point of the sum: adding a *third* class moved the answer **up**, and not because three beats two - the ITC fit simply found a tighter best site (3.1 against 13.1 uM). That tightest term alone carries **85%** of the ITC sum.

So a sum over sites is robust to missing a weak class and completely exposed to getting the strong one wrong. One laboratory, one preparation, two instruments, still 3.2x apart.

The saturation regime

The trace limit in step 4 is an approximation. The exact statement, straight from step 3, is

```
K(C_w) = \frac{K^0}{1 + C_w/K_d}
```

At `Cw = Kd` you are exactly 0.301 log (a factor of 2) below the plateau. This is not academic: it is precisely the basis on which Qin 2024 reports its specific-binding column, as `K_spec = C_max / (2 K_d)`, the ratio **at half-saturation**. Comparing that column with a converted Kd without adding 0.301 compares two different points on the same curve.

For PFOA on L-FABP, Kd = 0.099 uM:

Free Cw	log K
0 (plateau)	5.85
0.099 uM	5.55
1 uM	4.81

A full log unit lost between trace and a typical bioassay concentration.

The reverse direction is impossible

Algebraically the reverse is trivial - the same equation rearranged:

```
K_d = 10^{\,\log_{10}(10^9/M) \,+\, \log n \,-\, \log K}
```

In practice **zero** of the 207 partition-coefficient rows in our set can take it, and that zero is structural rather than a gap in our extraction.

The trace-limit measurement is

```
K^0 = \frac{10^9 \, n}{M \, K_d}
```

which determines only the **ratio** `n / Kd`. It cannot separate them, in principle, however well the experiment is done. A study that works in the trace regime - which is what makes its coefficient a constant at all - has thrown away the information needed to recover a Kd. To get one you need an experiment that reaches saturation, and then you are measuring a Kd directly and do not need the conversion.

This is worth saying plainly to anyone who asks for it: the forward and reverse directions are not symmetric, and no amount of extra extraction work will make them so.

What we tested: converted against measured

33 pairs, matched on compound, protein **and** species, never pooled across studies. Every converted value comes from a paper that supplied both the affinity and the stoichiometry; every measured value is Fischer 2024, the only source reporting human albumin partition coefficients on a mass basis.

Converted from	Pairs	Median residual	Where its n came from
Chen 2025	22	**-0.80**	single-site Langmuir, n fixed at 1
Wu 2024	4	-4.18	fluorescence double-log slope
Peng 2024	3	+0.13	fluorescence double-log slope
Chen & Guo 2009	2	-0.53	fluorescence double-log slope
Moro 2022	2	+0.28	ITC, n fitted from the isotherm

Residual = converted minus measured, log units. 27 of 33 are negative.

The one well-powered comparison is 0.80 log low with a consistent sign. 10^0.80 = 6.3, so closing it needs about six sites where a single-site fit allowed one. Maso 2021 co-crystallised **four PFOA per human serum albumin**, with one high-affinity and three low-affinity sites in solution - independent, structural, and of the right order.

The remaining 11 pairs are too few to read as a trend. They do establish one negative: three papers taking n off a fluorescence double-logarithmic slope land at -4.18, -0.53 and +0.13, so that slope is not a site count. Wu 2024's problem is not even its n - its association

constants sit three to four orders below every other measurement of the same compound on albumin.

The obvious explanation, tested and rejected

The natural story is non-specific binding: a Kd describes one saturable site, a measured coefficient also counts whatever sticks to the rest of the protein, so the converted value should be low by the non-specific share.

Qin 2024 is the only source in our set reporting the two terms **separately**, for bovine albumin, fitted to the same isotherm. At the trace limit:

Compound	nCF	Specific (plateau)	Non-specific	Total	Specific share
PFBA	3	2.61	1.83	2.67	86%
HFPO-DA	4	3.48	2.04	3.49	97%
PFHxA	5	3.43	2.56	3.48	88%
PFHpA	6	4.29	3.33	4.33	90%
PFHxS	6	5.19	3.44	5.19	98%
PFOA	7	4.75	3.75	4.79	91%
PFOS	8	5.44	4.04	5.45	96%

Dropping the non-specific term costs **0.07 log at most**. The gap to explain is 0.80. Non-specific binding is twelve times too small, and the sign of its chain-length trend is wrong too.

Obvious in hindsight: the specific term is saturable and therefore **largest** at trace, which is exactly the limit a converted Kd describes. Non-specific partitioning takes over higher up the isotherm, where neither number applies. The missing capacity is extra **sites**, not extra surface.

How big is 0.80 log? Five error scales

A residual means nothing without a yardstick. All five of these come from the same database, and the last one changes how the other four read.

What is being compared	Disagreement	Fold	n
One lab, same compound, two albumin batches	0.19 log	1.5x	10 pairs
Within-study chain-length line, leave-one-out	0.44 log	2.8x	12 groups
Two labs, same **measured** log K	0.59 log	3.9x	46 pairs

What is being compared	Disagreement	Fold	n
Kd -> K with n assumed = 1 (Chen 2025)	0.80 log	6.3x	22 pairs
Two labs, the same albumin Kd, same method	**1.45 log**	**28x**	4 compounds

The last row is the input to the conversion, and it is the largest number in the table.

The detail. Chen 2025 and Gao 2019 both measure PFAS on human serum albumin by equilibrium dialysis - the same technique, the same protein, the same species:

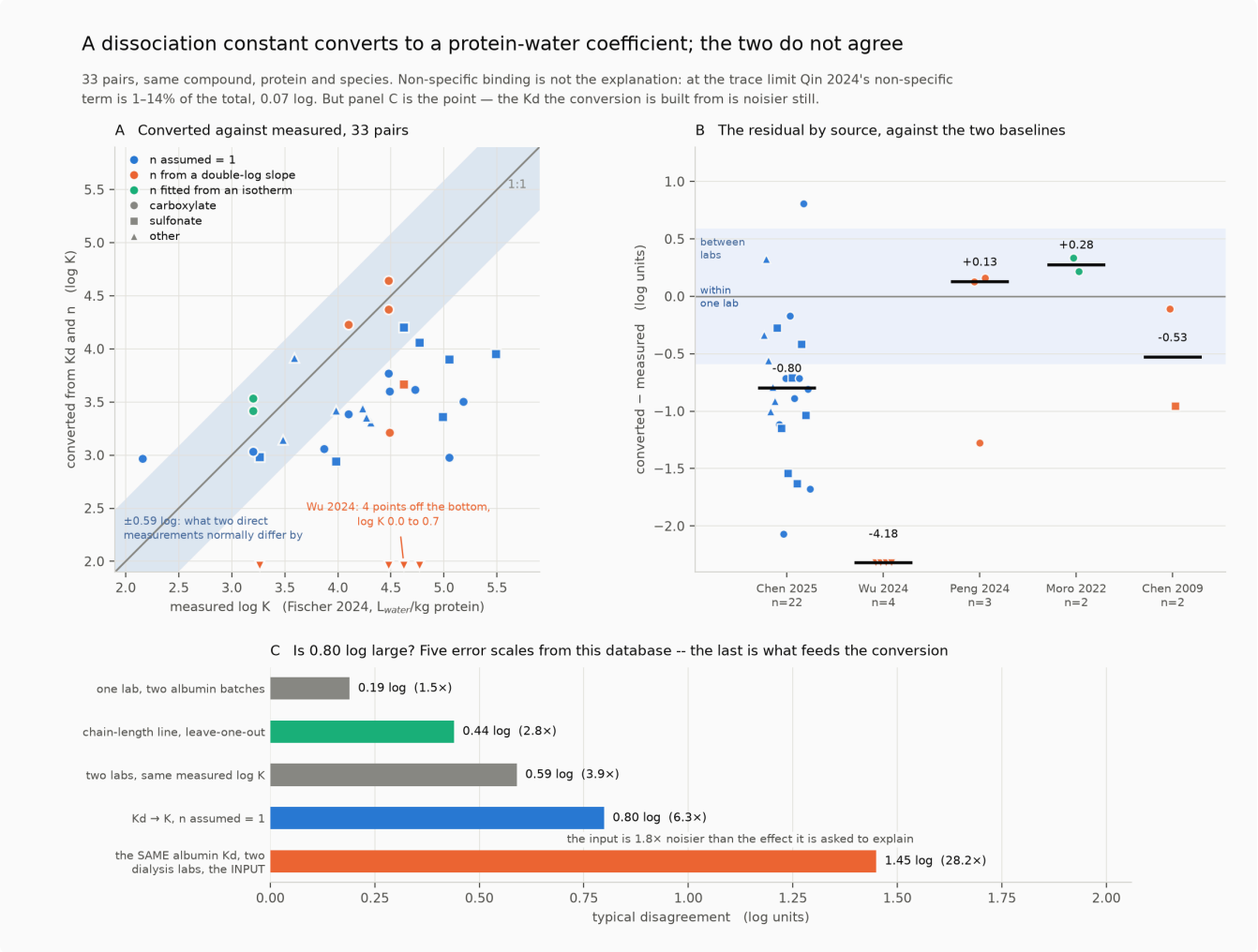
Compound	Chen 2025	Gao 2019	Gap
PFHxA	13.98 uM	195 uM	1.14 log (14x)
PFNA	3.75 uM	73 uM	1.29 log (19x)
PFOA	2.57 uM	115 uM	**1.65 log (45x)**
PFOS	0.94 uM	38 uM	1.61 log (40x)

Since `log K = log10(1e9/M) + log n - log Kd`, the converted value inherits that error **one for one**. The input is 1.8x noisier than the effect we set out to attribute. **An n wrong by six-fold is a second-order problem next to an input reproducible to 28x.**

Which way this cuts. It does *not* rescue the conversion - a one-sided offset across 22 compounds is not what noise produces. It says the offset is not worth attributing to any single cause while a larger, unattributed error sits underneath it.

And it stands alone as a finding. PFOA on human serum albumin is the most-measured protein-ligand pair in PFAS toxicokinetics, and two equilibrium-dialysis measurements of it differ by 45x. Any model that takes one of them as *the* value inherits that, and most models take one.

Note also the asymmetry in the table: a directly measured log K is reproducible to 3.9x between labs, while the Kd is reproducible to 28x. Where a mass-balance model can be built on either, it should be built on the measured coefficient.



Panel C is the one to read first: it answers "is 0.80 log big?" by putting it beside what agreement normally looks like in this database.

Affinity ranks sites, not compartments

A tighter Kd on liver FABP than on albumin does **not** mean the liver holds more PFAS. The mass a phase holds at trace is

```
\text{bound per litre of phase} = C_{protein} \cdot n \cdot \frac{C_w}{K_d}
```

so a comparison between compartments is a ratio of `C x n / Kd`, and the concentration term is the one **no binding paper reports**. Until we went and got it, our database had every Kd and no C, which is why this question could be asked and not answered.

The concentrations. Human hepatic FABP1 is **399 uM of cytosol** (211-760 across 61 livers, CV 24.8%), by surrogate-peptide LC-MS/MS. Plasma albumin is 527-752 uM (the 35-50 g/L interval).

These are **the same order**. The intuition that an intracellular binding protein is too scarce to be a compartment is simply wrong for L-FABP: per litre of cytosol it is 0.63x the molarity of albumin per litre of plasma.

The comparison, with no cross-study step. Jia 2022 measures both proteins by thermophoresis in one table, so five compounds carry both constants from one run:

Compound	L-FABP Kd (uM)	HSA Kd (uM)	Capacity ratio, per L cytosol
HFPO-TeA	1.22	179	**93x**
8:2 Cl-PFESA	5.39	31	3.6x
HFPO-DA	267.8	1156	2.7x
HFPO-TA	19.94	84	2.7x
PFEESA	3138	874	**0.2x**

Geometric mean 3.4x per litre of cytosol, 1.9x per litre of liver. Four of five favour liver; the spread across five compounds in one run is **527-fold**.

The ratio factorises exactly into a flat 0.63x concentration term and a compound-specific affinity term. So the concentration we went looking for turns out not to be what decides the answer - but that could only be established by having it. Had FABP1 come back at 1 uM, the affinity term would have been irrelevant.

What would overturn this. The ratio carries `n_HSA / n_LFABP` and thermophoresis reports neither. At Maso 2021's four albumin sites against one on L-FABP, every ratio falls four-fold and **only HFPO-TeA still favours liver**. The comparison also ignores interstitial albumin in the liver, which can only push the ratio down, and all of it is the trace limit.

Species and sex

Two questions with sharply different answers, and the contrast is the useful part.

Species: resolved, and smaller than the noise

Every row carries a species: 584 human, 170 bovine, 162 mouse, 99 rat, 31 rainbow trout, 8 porcine. Two sources compare species **within one laboratory**, which is the only comparison worth running given that two laboratories sit 1.45 log apart on the same constant.

Fischer 2024, human against bovine albumin, 27 shared compounds, one method: median difference **-0.02 log**, |median| 0.09 (1.2x), human higher in 12 of 27. That is no difference at all. The three largest are PFBS (+0.98), PFHpS (+0.61) and 5:3 FTCA (+0.58).

Starnes 2024, four species, seven compounds, one DSF run (Kd in mM):

Compound	Porcine	Rat	Human	Bovine	Spread
PFBA	1.28	1.90	2.64	3.06	2.4x
PFHxA	0.95	1.33	1.71	2.17	2.3x
PFOA	0.37	0.40	0.79	0.99	2.7x
PFBS	1.25	1.41	1.68	0.80	2.1x
PFOS	0.38	0.42	0.69	0.86	2.3x
HFPO-DA	2.12	1.64	1.57	2.55	1.6x
6:2 FTSA	0.52	0.33	0.41	1.39	4.2x

Median spread **0.36 log (2.3x)**. The order porcine < rat < human < bovine holds for 4 of 7 - an ordered effect, but a small one. 6:2 FTOH is a non-binder in all four species, which is the cleanest species agreement there is.

Put against the yardsticks, both species comparisons fall below the floor:

	log	Fold
Human vs bovine albumin, one lab	0.09	1.2x
One lab, two albumin batches	0.19	1.5x
Four mammals, one lab	0.36	2.3x
Two labs, same measured log K, same species	0.59	3.9x
Two labs, same albumin Kd, same method	1.45	28x

Changing the mammal moves a PFAS-albumin constant less than changing the laboratory does. Two consequences: borrowing an albumin constant across mammals costs less than borrowing one across papers within a species - the usual worry is backwards; and a 0.36 log species effect cannot be resolved by pooling literature values, because the pooling adds 0.59 to 1.45. It needs one lab measuring both species.

Where the species difference is real: the matrix, not the protein

Albumin is albumin. Plasma is not plasma. Qin 2024 puts human against rainbow trout **plasma** at a median -0.45 log over 16 compounds. But Qin also measures the composition - human plasma 42.25 mL/L protein and 4.46 mL/L lipid, trout 15.46 and 7.08 - and those two compositions alone confine the ratio to 0.30x-1.32x for *any* pair of coefficients. Five of eleven measured ratios fall outside that bound.

So the surviving species difference is in how much protein and lipid the fluid contains, plus something composition cannot explain - not in the protein's binding constant.

Sex: not resolved, and that is structural

There is no sex field, and 6 of 1,241 rows mention sex anywhere in free text, from 4 studies. This is not an extraction gap. A binding assay uses purified protein, and purified albumin has no sex, so no paper reports it.

That matters because the toxicokinetic side of this project shows enormous sex effects:

PFAS	Species	Half-life F/M	Clearance F/M	% from clearance
PFOA	rat	0.049	44.2	82.4%
PFNA	rat	0.053	20.6	95.4%
PFHxS	rat	0.059	20.5	94.6%
PFBA	rat	0.281	3.6	98.3%

Rat PFHxS eliminates 17x faster in females, and 94.6% of that is **clearance** - renal transport under androgen control - not binding.

The conclusion for anyone extrapolating: binding is the part of PFAS toxicokinetics that travels between species and sexes most safely. Albumin constants move less than 2.5x across four mammals and cannot move at all with sex. Transport is the part that does not travel. A model that gets species or sex extrapolation wrong is almost never wrong in its binding term.

Practical rules

1. **Convert only within one paper.** It must supply the affinity and the stoichiometry together. Taking a Kd from one study and a coefficient from another gives a reconciling n spanning 752-fold, ordered by chain length - a diagnosis of why the rule exists, not a method.
2. **Carry the protein's own mass.** `log10(1e9/M)`, not 4.1774 by habit.
3. **State the basis on every output.** Per kg and per litre differ by 0.135 log. A number without its basis is unusable.
4. **Label a converted value what it is**: the specific term, not the partition coefficient. Above about C8 the specific term is the smaller of the two contributions.
5. **Never convert in reverse.** It is structurally impossible, not merely unextracted.

6. **Prefer a measured coefficient where one exists.** It is reproducible to 3.9x between labs; a Kd is reproducible to 28x.
7. **If you only need an interpolated log K**, a within-study chain-length line (one study, one head group, one protein, inside the measured range) gives 0.44 log and is unbiased by construction. It cannot tell you anything about saturation, because it contains no Kd.

Conversion inventory

	Rows
Albumin rows in the database	543
Convertible forward (paper gave n)	82
Convertible in reverse	0
FABP and PPAR rows	55
...of those, convertible	5
Blocked by no stoichiometry	18
Blocked by unstated construct mass (PPARs)	7

Sources

The ten sources supplying the most measured constants against a named single protein, which is the scope of this document:

Rank	Source	Measured rows	Compounds	Quantity
1	Fischer 2024	83	29	log D - HSA, BSA, globulin
2	Chen 2025	57	57	Kd - HSA
3	Muensterman 2026	56	56	log D - HSA, biomimetic
4	Gao 2025	42	10	Ka - BSA, NaCl series
5	Qin 2024 (plasma)	32	16	log D - BSA, by concentration
6	Starnes 2024	32	8	Kd - albumin, 4 species
7	Starnes 2025	28	28	Ka - HSA, DSF

Rank	Source	Measured rows	Compounds	Quantity
8	Allendorf 2019	24	14	log D - BSA, dialysis
9	Jia 2022	13	9	Kd - hL-FABP + HSA, MST
10	Bischel 2011	11	11	log D - BSA

Findings here that rest on papers outside that set, flagged so nobody is surprised later:

- **Gao 2019** (ES&T 53:6529) - the 1.45 log albumin Kd discrepancy, the single most consequential number in this document.
- **Khazaee 2021** (Toxics 9:45) - the L-FABP equilibrium-dialysis constants and the PPAR rows.
- **Woodcroft 2010** (ETC 29:1669) - the multi-site worked example.
- **Maso 2021** - four PFOA per albumin by co-crystallography.
- **Wen & Isoherranen 2026** (JPET 393:104309) - hepatic FABP1 at 399 uM, the concentration that made the capacity comparison possible.
- **Moro 2022, Peng 2024, Wu 2024, Chen & Guo 2009** - the 11 non-Chen converted pairs.

Reproducing this

All figures come from scripts in the repository, each of which prints its own derivation: `kd_k_interconversion.py` (the conversion and its row inventory), `logk_converted_vs_measured.py` (the 33 pairs and the five error scales), `lfabp_capacity_vs_albumin.py` (the capacity comparison), `make_logk_conversion_figure.py` (the figure), binding_species_sex_resolution.py (species and sex).